

Jawaharlal Nehru Govt. Engineering College, Sundernagar, Distt. Mandi(H.P)

Department of Computer Science and Engineering

Program:	B.Tech	Semester:	3rd
Subject Code:	CSPC-311	Subject Title:	Data Structure and Algorithm
MST No.:	1st	Course Coordinator:	Rahul Pal Singh
Maximum Marks:	50	Time Duration:	2hrs.
Date of MST1:	03-10-2024 (Evening)	Roll Number:	

Note: All Questions are Compulsory

Q. No.	Questions	CO/Marks
Q1.	Implement the algorithm to reverse the singly linked list. What is the complexity of this algorithm? Demonstrate step by step reversal of the following linked list. START → 6 → 1 → 9 → 3 → 8 → NULL	CO2, L3 4+3+3=10
Q2.	a. What do you mean by asymptotic notations? Explain Big Oh Notation in detail. Show that $4n^4 + n^3 + n = O(n^4)$ b. Algorithm A has time complexity of $24 * n^2$ while algorithm B has time complexity of $64 * n * \log(n)$. Which algorithm is better to solve the given problem? Also find the crossover point.	CO1, L3 7.5+7.5=15
Q3.	What is an array? How is it represented in a memory? Briefly explain algorithms to: a) Declare, initialise, and search the array with following data values: 3, 8, 12, 34, 76, 2, 13, 32. b) Traverse an array. c) Invert the elements of an array.	CO2, L3 3+3+4=10
Q4.	Find the running time of following C code snippet in terms on input n: <pre>for (a=1; a<n; a*=2) { for (b=n; b>0; b/=2) { for (c=b; c<n; c+=2) { sum += (a + b*c); } } }</pre>	CO1 6
5.	a. Explain Heap as a data structure. Write an algorithm to insert data in a min heap. b. Given the input data: 25, 34, 32, 21, 67, 54, 12, 2, 98, 3, 11, 20 i. Create a min heap ii. Create a max heap	CO3 5+5=10

Course Outcomes(CO):

Compare the performance and analyse the complexity of various data structures

Able to design and implement Linear data structures like linked lists, queues, stacks, etc.

Able to design and implement Non-Linear data structures i.e. binary search tree, heap trees, etc

Able to perform operations on graph data structure

Learn & Compare algorithmic techniques to implement various sorting algorithms

Understand and apply the concept of hashing and hash table operations.

Level Number	L1	L2	L3	L4	L5	L6
Level Name	Remember	Understand	Apply	Analyze	Evaluate	Create